

RESEARCH & EDUCATION TOOL / NOT A MEDICAL DEVICE

TIPS ICE Planner

User Manual

A pre-procedure TIPS planning tool that, from a contrast-enhanced CT, helps you rehearse where along the hepatic vein — and in what direction — a puncture would reach the portal vein.

This app is a **research, education and self-training tool**. It is **not a certified medical device** and is **not** intra-procedural navigation (it works from the pre-procedure CT, and the liver and vessels move with respiration and the needle itself during the real procedure, so the on-screen geometry will not match exactly). All numbers and paths shown are **descriptive reference information only** — they do not imply fewer punctures, reduced radiation, or that any particular needle or path is safer. **The operator makes all final clinical decisions.**

Author: M. Yamamoto (IR physician, Japan)

1. Open a patient

On launch you see a list of studies. Import a DICOM folder (e.g. from a CD-R) with “**Import DICOM folder...**” and it stays in this list for next time.

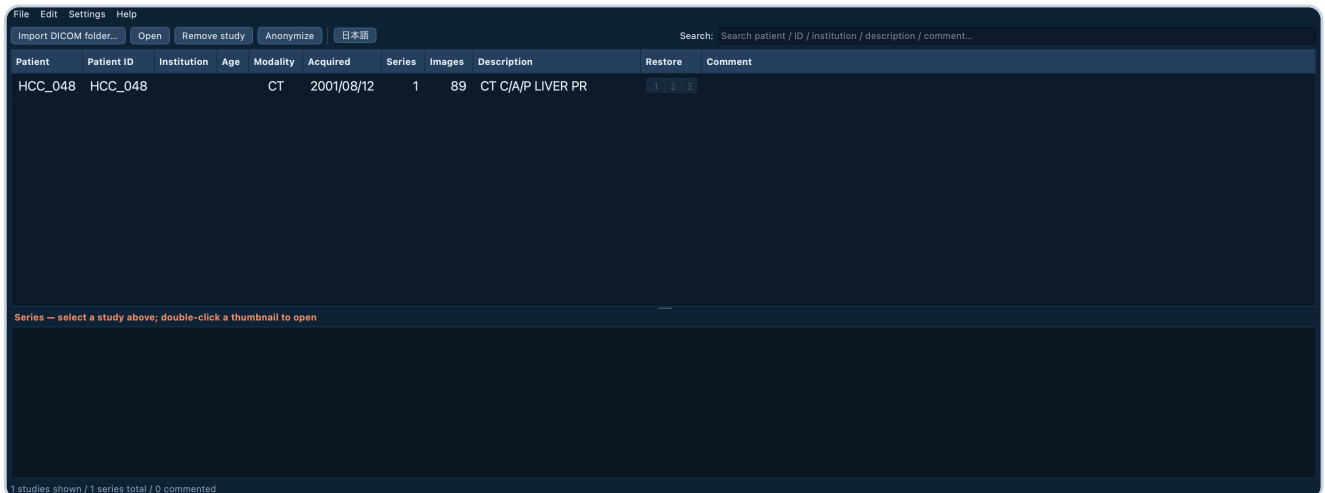


Figure 1 — Patient list. Select a row and click Open (or double-click a thumbnail). Shown here with a public TCIA dataset (CC BY 4.0)

- **Search:** filter studies by patient name, ID, **institution**, description or comment.
- **Institution column:** shows the DICOM institution name for each study.
- **Anonymize:** temporarily hides name / ID / age for screen sharing or teaching (the underlying data is unchanged).
- **Comment:** click the cell once to edit it in place — handy for teaching notes or a quick summary of the case.
- **Restore column:** recalls a saved working state (slot 1, 2 or 3) directly from that patient's row — it always targets that patient regardless of which row is selected. Empty slots are shown dimmed and can't be clicked. See “6. Save, restore and undo” below.
- **Remove study:** removes it from the list only — the DICOM files on disk are kept.

The CT you open here becomes the working volume for Step 1 and Step 2 below. You can always re-open the same patient later from this list.

2. Step 1 — ICE setup

The “1. ICE setup” tab builds the geometry of a virtual side-firing ICE probe sitting inside the IVC (inferior vena cava): which direction it points, and what cross-section it would show. All the controls live in a single strip at the bottom of the window.

1. On the **Axial pane**, click **≥2 points along the IVC**. The points you click, joined together, become the probe's path through the IVC (you can drag any point afterward to fine-tune it).
2. Choose **Insertion** (Femoral / Jugular) from the menu bar under “**Settings**” → “**Default insertion route**”. Since this is usually fixed per institution rather than changed case-by-case, it has been moved out of the per-case controls and into Settings.
3. Aim the fan with the bottom control strip (below).



Figure 2 — Overall layout. Four panes (Axial / Coronal / Sagittal / ICE view) plus the 3D linkage panel on the right, with the control strip along the bottom

The bottom strip uses **Handle** mode: you drag a picture of the probe handle to steer the ICE view.

Control (Handle mode)	What it does
Drag the body up/down	Rotate θ (spins the ICE fan around the IVC axis — matches turning the real catheter)
Drag the body left/right	Push/Pull (slides the probe along the IVC path toward the foot/head)
Drag the two thumbwheels up/down	Deflect A-P · L-R (simulates deflecting the tip anterior/posterior and left/right)

Other controls	What it does
Zero deflect	Resets both A-P and L-R deflection to 0° at once
Roll / Roll 0	Freely rotates the ICE image itself for a more convenient view (does not change the underlying geometry) / resets it to 0°
Flip ICE L/R	Mirrors the ICE image left–right
Liver / Liver: Haze · Surface / Opacity	Overlays a translucent liver “ghost” to help judge position; toggle haze/surface style and adjust opacity

The ICE view is rendered as **plain CT** (no ultrasound speckle or texture is simulated). It exists to quickly show “which direction, and where, am I looking right now.” Overlaying the liver ghost makes it easy to see roughly where in the liver the fan and probe are pointing. The real procedure will not match this angle-for-angle, but it helps build a mental picture beforehand.

3. Step 2 — Puncture planning

The “2. Needle” tab lets you set Entry (the hepatic-vein puncture point) and Target (the portal-vein aim point), and review the straight-line path between them. Use the “Set:” row at the bottom to choose which one you're placing.

1. With **1•Entry (puncture)** selected, click on any pane.
2. With **2•Target (portal)** selected, click the portal-vein position you're aiming for.

The straight Entry→Target path is drawn on all four panes and in 3D. A dedicated text readout under the ICE view shows the straight-line distance and color-coded status:

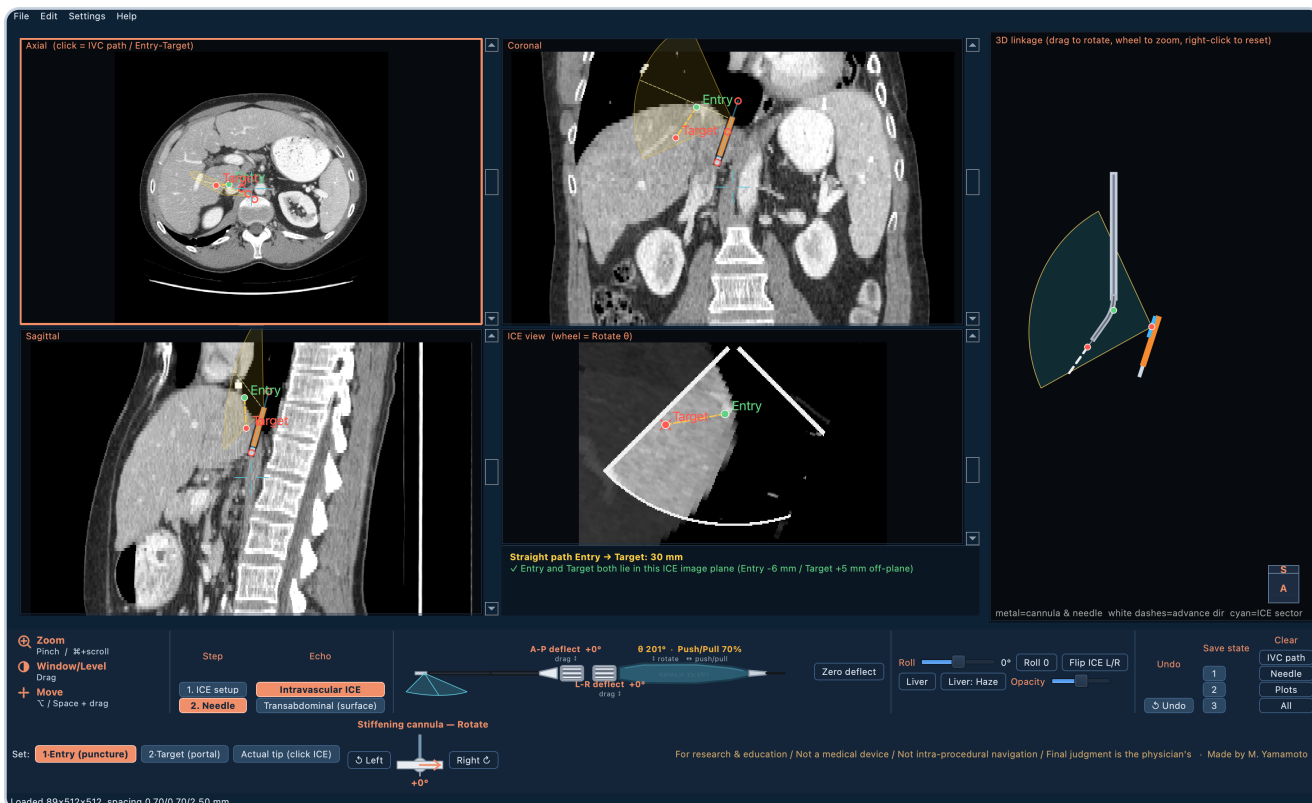


Figure 3 — Entry and Target set. The readout under the ICE view shows the straight-line distance and whether both points lie in the current plane

- **Yellow:** the straight-line Entry→Target distance.
- **Green / Orange:** whether Entry and Target both lie in the current ICE plane. If not, it shows the off-plane distance (mm) and how many degrees to rotate θ to bring the whole path into view.

Curved-needle models (RUPS / Colapinto) and the “Plot” rehearsal mode are temporarily disabled pending redesign. The straight-line path is the current foundation, with curved models planned to return in a future update.

4. Recording the actual needle position

Turning on “**Actual tip (click ICE)**” makes a click on the ICE image record “where the needle tip is actually seen right now.” Entry stays fixed; a translucent needle shape connects it to the point you clicked (drag the tip afterward to fine-tune it).

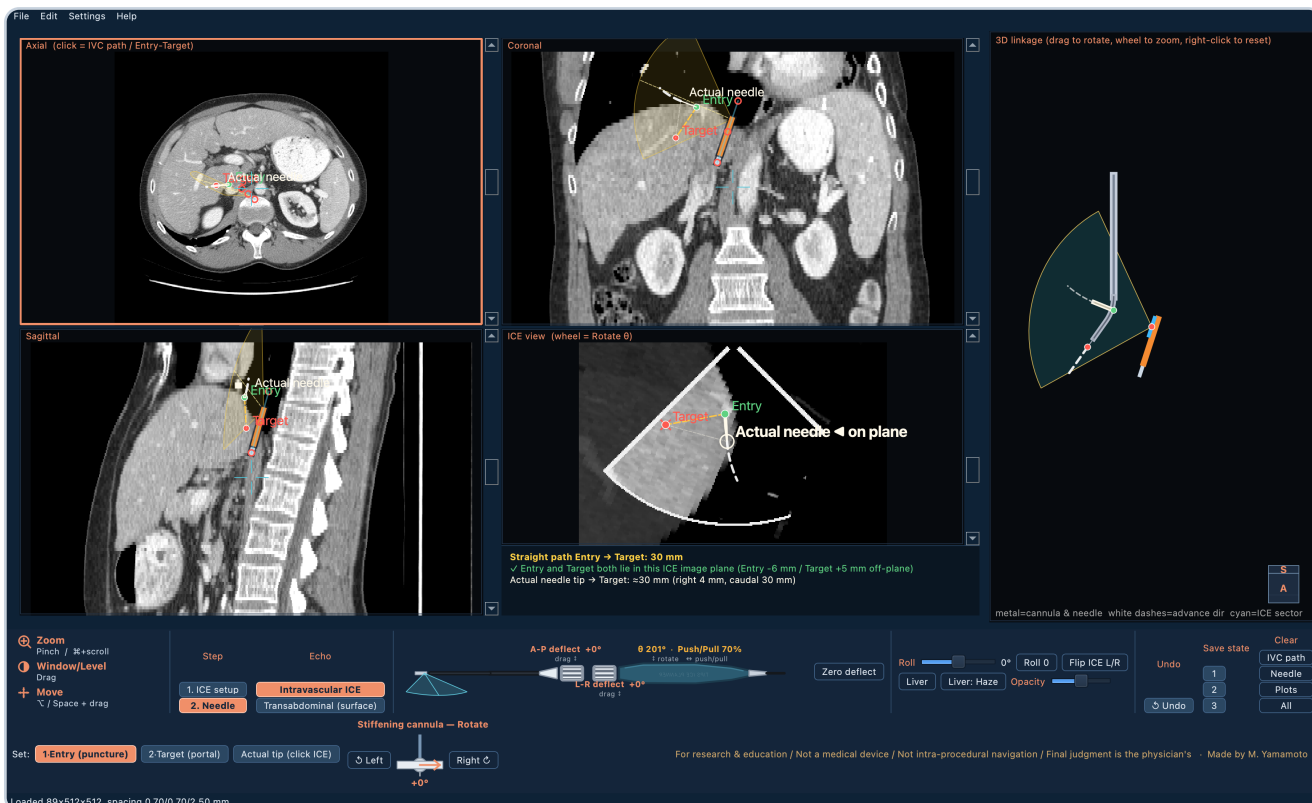


Figure 4 — Actual tip set. The readout under the ICE view shows the distance from the actual tip to Target, broken down by left/right and cranial/caudal offset

About the emphasis. The actual-needle shape (Entry → tip) is drawn thicker and darker, with a ring at the tip and a bold “◀ on slice / on plane” label, **only when it truly lies in the CT slice or ICE plane you're currently viewing**. Rotate θ or scroll to a different slice and, once the true position falls outside the view, the needle fades and the label disappears. This is meant to answer, at a glance: “is this needle actually visible in what I'm looking at right now?”

A short ($\approx 2\text{cm}$) dotted extension is also drawn from the tip, animated flowing forward. It represents the assumed continuation if a Colapinto-like curvature is applied. The **Rotate left / Rotate right** buttons twist this assumed curvature to show which way the tip would bend if you twisted the stiffening cannula by hand.

All of this is **descriptive reference information only** — not a recommended path, not an optimization, and not a safety guarantee.

5. Transabdominal (surface) echo mode

Switching “**Echo: Intravascular ICE ↔ Transabdominal (surface)**” at the top simulates a convex probe placed on the skin. Click on the skin in any of Axial / Coronal / Sagittal to snap the probe onto the body surface; drag to slide it along the skin.

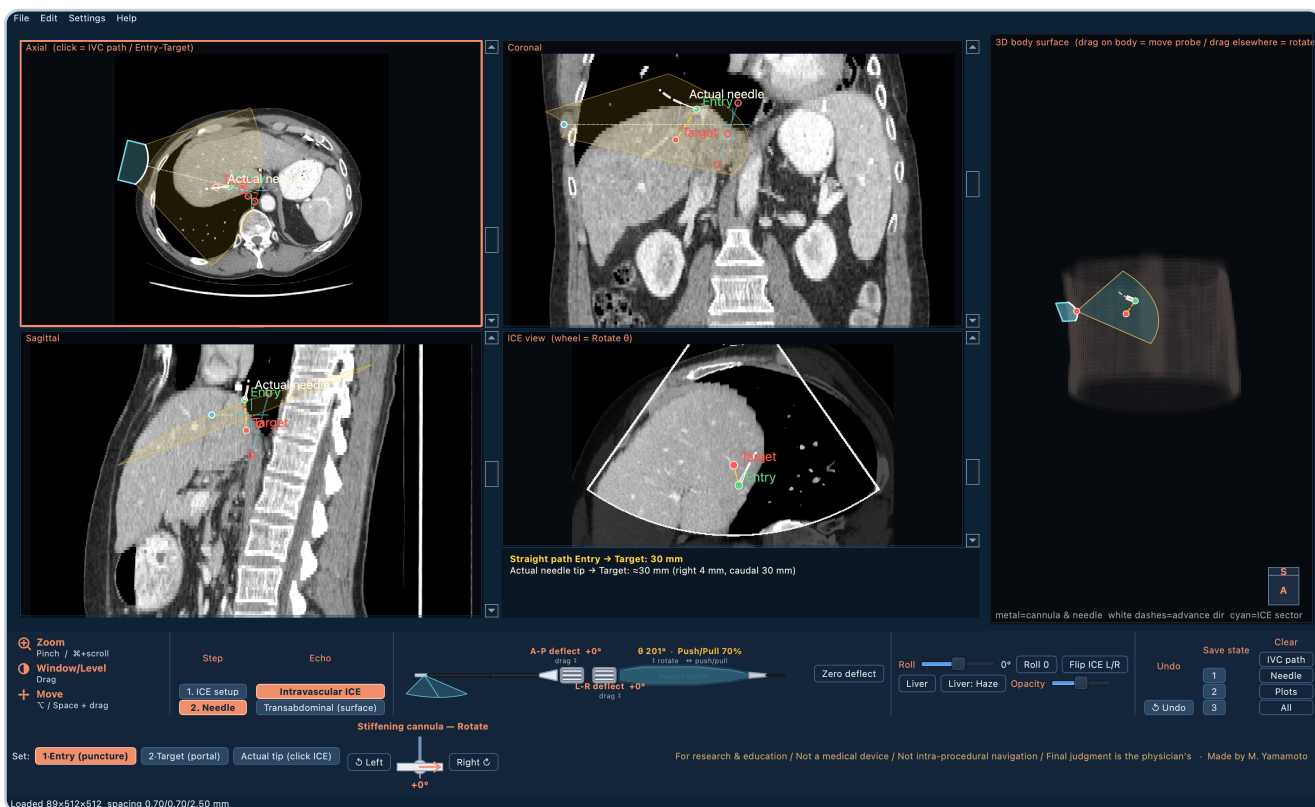


Figure 5 — Transabdominal mode. The probe and fan ghost are projected onto all three cross-sections; the 3D panel shows the outer body-surface shell

In this mode the θ /deflection labels change to “Rotate / Tilt (in-plane) / Rock (out-of-plane)”, and probe Push/Pull is hidden. This is a **static CT preview**, not live ultrasound.

6. Save, restore and undo

The bottom control strip also has controls for saving, restoring, and undoing your work.

Control	What it does
Save state (1 / 2 / 3)	Saves the current working state (IVC path, Entry/Target, actual tip, view settings, etc.) into one of three slots dedicated to this patient. Up to three states can be kept per patient.
Restore	To bring a saved state back, use the Restore button (1, 2 or 3) on that patient's row in the patient list — not from within the viewer itself.
Undo	Steps back one action — the last click/point placement, or a Clear / zero-deflect / Roll-0 / insertion-route change (mid-drag states are not covered).
Clear	Removes just the IVC path, the needle (Entry/Target/actual tip), plots, or everything — individually.

Closing the app asks whether to save the current patient's working state. Choosing “Yes” saves it to the first empty slot automatically and pops up which slot number it used.

7. Menu bar, the 3D linkage panel, and common gestures

The top menu bar is organized as follows.

Menu	Key items
File	Patient list / Open DICOM
Edit	Undo / Clear IVC path / Clear needle / Clear plots / Clear all
Settings	Default insertion route / Switch language (English ↔ Japanese)
Help	User manual (PDF) / Tip of the day / FAQ / About / Check for updates / Send feedback / Support development

The “3D linkage” panel on the right lets you inspect the ICE probe, fan, liver and actual needle in three dimensions.

Action	Effect
Drag	Rotate the viewpoint
Wheel / pinch	Zoom, centered on the cursor
Right-click	Reset zoom / viewpoint

Every image pane (Axial / Coronal / Sagittal / ICE) shares the following controls.

Action	Effect
Pinch /  +scroll	Zoom, centered on the cursor
Left-drag	Adjust window level / width (brightness & contrast)
Right-drag / wheel	Zoom, centered on the cursor
Right-click (click only, no drag)	Reset zoom & position
 Option /  + drag, or press-and-drag the scroll wheel (middle-drag)	Pan the image

Disclaimer. This app is a prototype for research, education and self-training only. It is not a certified medical device, is not intended to diagnose or treat disease, and is not intra-procedural navigation. All numbers, paths and emphasis shown are descriptive reference information only; they do not imply that any particular needle, path or technique is safe or recommended. Final clinical judgment always rests with the operator.